

4 CORE 200 Reading | Module 4 | Causal Graphs

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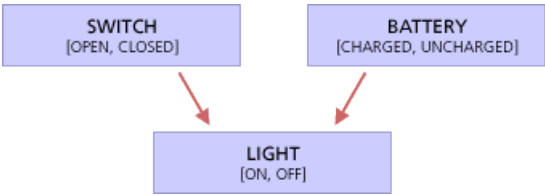
4.0.1 Module 4 | Introduction

In previous modules, we built up an account of what it is for one variable X to be a direct cause of another Y. Informally, if we can intervene to change X while holding everything else fixed, and still produce a change in Y, even the chances of Y, then X is a direct cause of Y. The formal definition of direct cause is even more complicated, involving test pairs and response structures. What we do in this module is give you a compact, pictorial way to represent the causal relations, direct and indirect, among a system of variables.

Causal relationships between variables are often represented by arrow diagrams. We draw an arrow from a variable X to a variable Y if X is a direct cause of Y (relative to the set of variables in the diagram). So, for example, the claim that water temperature influences the height of the water in a glass (by making the water expand or contract) can be represented by the following diagram, where the boxes are variables with the possible values they might take on in square brackets.



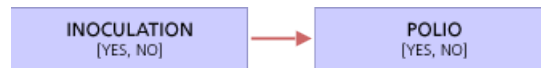
The causal relations between the state of a light bulb, a light switch and a battery can be represented by a causal graph involving three variables:



and the relations between the refrigerator door, the light switch, and the refrigerator light by another:



We use the same kind of diagram no matter whether the cause tends to prevent the effect or to bring the effect about. So we would represent the claim that an inoculation (with the Salk polio vaccine) prevents Polio by the following graph.



We use other means to indicate whether the causal factor tends to bring about or prevent the effect; for example, we might place a plus or minus sign next to the arrow :



In this case, the minus sign indicates that bringing about the cause (Inoculation = yes) tends to make it LESS likely that the effect will occur (Polio = yes). In other cases, where the variables have values that increase or decrease, a plus sign might indicate that by increasing the cause, e.g., moving humidity from low to medium to high, we make it more likely that the effect will increase as well, e.g., sweat produced will be more likely to be higher.

This module explains how causal graphs represent, in a qualitative way, the causal relations among a set of variables. It also introduces and defines features of causal graphs that will be crucial in understanding the connection between causal systems and statistical data, for example: common cause, direct vs. indirect causation, common effect, and more.

The module stresses four basic skills and/or concepts:

4.0.2 Concepts and Skills Covered in Module 1.4

1. Elements of Causal Graphs
2. Causal Graphs and Response Structures
3. Representing Different Varieties of Causation as Graphs
4. Translating Text into Causal Graphs

4.1 The Elements of Causal Graphs

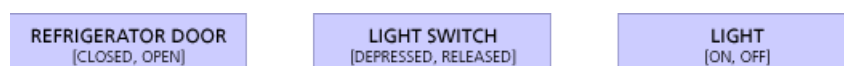
4.1.1 Module 4 | Variables

Causal graphs represent the causal relations in a causal system. Specifically, causal graphs involve:

1. a set of variables, and
2. a set of directed edges that connect the variables.

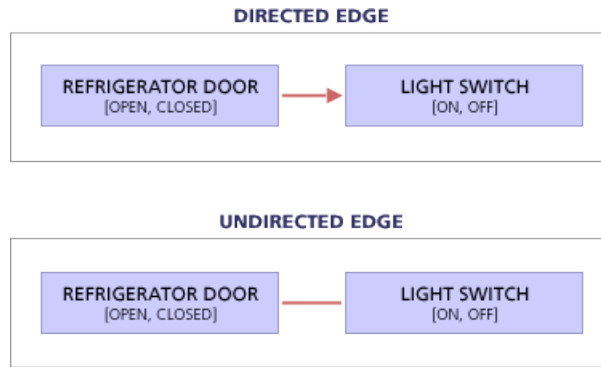
Variables were introduced in the module on Causation: Preliminaries, so we only give a brief overview here. The values of variables are properties of an individual, e.g., the hair color of a person, the population of a country. The set of values for a variable must be both exclusive and exhaustive. A set of values is exclusive if no individual can have more than one value. A set of values is exhaustive if every individual has one of the values. For example, consider the variable Hair Type with values: [Red, Blond, Short], that are neither exclusive nor exhaustive. Someone can have hair that is both blond and short, so the set is not exclusive, and someone can have hair that is black and long, in which case no value from the set applies so it is not exhaustive.

In a causal graph, we represent each variable as a box with the variable's name and possible values (though we will sometimes leave out the variable values). So, if our variables are REFRIGERATOR DOOR [Closed, Open], LIGHT SWITCH [Depressed, Released], and LIGHT [Off, On], then we need to have a box for each variable:



4.1.2 Module 4 | Directed Edges

A directed edge in a causal graph is an arrow, where the *head* of the arrow points to the effect variable and the *tail* comes from the cause variable. We say it is a *directed edge* to distinguish it from an *undirected edge* :



We include a *directed edge* from a variable X to a variable Y in the causal graph that represents a set of variables S if and only if X is a *direct cause* of Y relative to S .

We gave an account of “cause” in the Causation Among Variables and Indeterministic Causation modules. The key concepts involve *causal assignments*, *response structures*, and a *test pair of causal assignments*. If you need to review these ideas, you can review these modules at any time. Here again are the key definitions:

Test Pair of Causal Assignments If two causal assignments $C1$ and $C2$ are identical except for the values assigned to variable X , then $C1$ and $C2$ are a *test pair of causal assignments* for X .

Direct Cause If, in a system of variables S there are any test pair of causal assignments for X in which there is a difference in the effect Y , then X is a *direct cause* of Y relative to S .

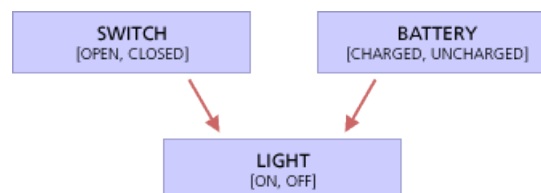
In the module on Indeterministic Causation, we explained how this definition still covered cases of indeterministic causation in which the “difference in the effect” amounts to a change in the probability it will occur.

Let’s examine how a causal graph represents the causal relations in a system with a few simple examples.

Example (A) Example 1: Switches and Lights

Consider a causal system among the variables BATTERY, SWITCH and LIGHT.

The causal graph of this system is as follows:



Notice that there is *no* arrow from the switch to the battery, nor from the battery to the switch, even though they are physically connected by wire on the circuit. Why? Because the state of the battery has no causal influence on the state of the switch, nor does the state of the switch (in this idealized example) have any direct influence on the state of the battery. Intervening to change the state of the battery will not affect the state of the switch, even though the two are physically connected.

Notice second that there *is* an arrow from the switch to the light bulb, even though, when the battery is uncharged, changing the switch from open to closed (or from closed to open) will not change the state of the light bulb (it will stay off).

There is an arrow from the switch to the light bulb because there is *some* state of the battery, namely when it is charged, for which changing the causal assignment of the switch does change the state of the light.

Example (B) Example 2: Lighting a Match

The definition of direct cause highlights the role of the other variables we are considering. Sometimes, whether an edge should be included in a graph depends on those other variables. Consider the causal graph for the process

of lighting a match:



There is no edge that goes directly from STRIKE MATCH to MATCH LIGHTS. Why? Apply the definition above to this case:

Is there a test pair of causal assignments that differ only by the value assigned to STRIKE MATCH that make a difference to the effect MATCH LIGHTS? No. If the TIP TEMPERATURE is above 350, then the match will light whether we strike it or not. If the TIP TEMPERATURE is below 350, then the match will not light whether we strike it or not. Thus there are no test pairs for STRIKE MATCH that make a difference to MATCH LIGHTS, even to its probability.

4.1.3 Module 4 | Deterministic Causation: The Malaria Example

Consider the case of malaria again. The variables in the first causal system are:

Variables For The Malaria Case	
Variable	Value
BITTEN (Was bitten by an infected mosquito)	[True, False]
INOCULATED	[True, False]
HAS GENE (Has the sickle cell gene)	[True, False]
DRINKER (Drinks gin and tonics regularly)	[True, False]
MALARIA (Gets malaria)	[True, False]

The response structure for malaria was given by the following table in the module on Variable Causation:

Response Structure For The Malaria Case					
Assignment	Variable 1: BITTEN	Variable 2: INOCULATED	Variable 3: HAS GENE	Variable 4: DRINKER	Effect: MALARIA
1	True	True	True	True	False
2	True	True	True	False	False
3	True	True	False	True	False
4	True	True	False	False	False
5	True	False	True	True	False
6	True	False	True	False	False
7	True	False	False	True	True
8	True	False	False	False	True
9	False	True	True	True	False
10	False	True	True	False	False
11	False	True	False	True	False
12	False	True	False	False	False
13	False	False	True	True	False
14	False	False	True	False	False
15	False	False	False	True	False
16	False	False	False	False	False

Should there be a direct arrow from the variable BITTEN to the variable MALARIA in the causal graph representing

this system? How do you answer this question? Not by guessing, not by common sense, but by applying the definition for direct cause to the response structure above.

The definition requires that there is at least one test pair of causal assignments for BITTEN that make a difference to MALARIA. So to apply the definition, first locate the test pairs, and then check to see if there are any in which the value of MALARIA is different.

In this case, those pairs are: 1 and 9, 2 and 10, 3 and 11, 4 and 12, 5 and 13, 6 and 14, 7 and 15, and 8 and 16. Is the value of MALARIA different across any of these pairs? Put another way, is the value of MALARIA different across causal assignments 1 and 9? Is it different across causal assignments 2 and 10?

The answer is yes. In causal assignments 8 and 16, where the variables besides BITTEN take on the values:

Variable Values	
Variable	Value
INOCULATED	False
HAS GENE	False
DRINKER	False

then changing the value of BITTEN changes the effect MALARIA. So BITTEN is a direct cause of MALARIA.

4.1.4 Module 4 | Indeterministic Causation: The Cell Phone Example

Consider the case of the Cell Phone again. In the full, deterministic causal system, there are three variables:

Variables For The Cell Phone System	
Variable	Value
CALL PLACED	[Send, End]
IN RANGE OF TOWER	[Yes, No]
CONNECTED	[Yes, No]

The response structure for CONNECTED is as follows:

Deterministic Response Structure For Connected			
Assignment	CALL PLACED	IN RANGE OF TOWER	CONNECTED
1	Send	Yes	Yes
2	Send	No	No
3	End	Yes	No
4	End	No	No

Would the causal graph among these three variables have a directed edge from CALL PLACED to CONNECTED? Yes, because there is a test pair for CALL PLACED that makes a difference to CONNECTED.

In causal assignments 1 and 3, where IN RANGE OF TOWER is assigned “Yes,” then changing CALL PLACED from End to Send always changes the value of CONNECTED from No to Yes.

Now consider the pseudo-indeterministic system involving just these variables:

Variables For The Pseudo-Indeterministic System

Variable	Value
CALL PLACED	[Send, End]
CONNECTED	[Yes, No]

Would the causal graph among these two variables still have a direct edge from CALL PLACED to CONNECTED? How do we answer the question in this case, where the causation is indeterministic? First we write out the indeterministic response structure, and then apply the definition of indeterministic causation for variables.

Indeterministic Response Structure For Connected

Assignment	CALL PLACED	CONNECTED = Yes	CONNECTED = No
1	Send	50%	50%
2	End	0%	100%

Here is the definition of indeterministic causation for variables we gave in the module on Indeterministic Causation

Direct (Indeterministic) Cause If, in a system of variables S , there are any test pairs of causal assignments for X in the response structure for Y across which the probability distribution over Y differs, then X is a *direct cause* of Y relative to S .

So we need to apply this definition to the indeterministic response structure above. Causal assignments 1 and 2 are a test pair for CALL PLACED, and there is indeed a difference in the probability over CONNECTED across these two assignments. So, by applying the definition to the indeterministic response structure, it is clear that CALL PLACED is a cause of CONNECTED.

In the cell phone example, we were implicitly assuming background conditions that include a functioning cell phone with a charged battery, etc. Let's consider whether or not the battery is charged as another variable, instead of a part of the background conditions. The system now includes the variable PHONE BATTERY CHARGED [Yes, No]. So now the full system is:

Cell Phone System With A New Variable

Variable	Value
CALL PLACED	[Send, End]
PHONE BATTERY CHARGED	[Yes, No]
IN RANGE OF TOWER	[Yes, No]
CONNECTED	[Yes, No]

Now the response structure for CONNECTED is as follows:

Response Structure For The Cell Phone System

Assignment	Variable 1: CALL PLACED	Variable 2: BATTERY CHARGED	Variable 3: IN RANGE OF TOWER	Effect: CONNECTED
1	Send	Yes	Yes	Yes
2	Send	Yes	No	No
3	Send	No	Yes	No
4	Send	No	No	No
5	End	Yes	Yes	No

Continued on next page

Response Structure For The Cell Phone System (Continued)

Assignment	Variable 1: CALL PLACED	Variable 2: BATTERY CHARGED	Variable 3: IN RANGE OF TOWER	Effect: CONNECTED
6	End	Yes	No	No
7	End	No	Yes	No
8	End	No	No	No

Now suppose we consider the pseudo-indeterministic system in which we cannot observe whether or not we are in range of the tower:

Pseudo-Indeterministic Cell Phone System

Variable	Value
CALL PLACED	[Send, End]
PHONE BATTERY CHARGED	[Yes, No]
CONNECTED	[Yes, No]

Is there still a directed edge from CALL PLACED to CONNECTED?

4.2 Representing Different Varieties of Causation

4.2.1 Module 4 | Common Causes

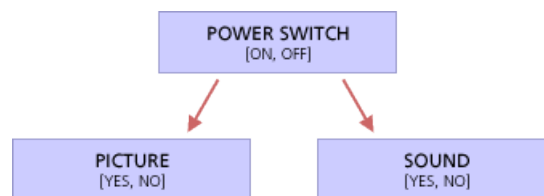
A variable C is a common cause of two or more other variables X and Y when C is a cause (direct or indirect) of both X and Y.

Consider the following three variables for TVs that function normally:

Variables For A Tv System

Variable	Value
SOUND	[Yes, No]
POWER SWITCH	[On, Off]
PICTURE	[Yes, No]

The causal graph for these variables is:



Here POWER SWITCH is a common cause of both PICTURE and SOUND because changing the state of POWER SWITCH changes both the value of PICTURE and the value of SOUND.

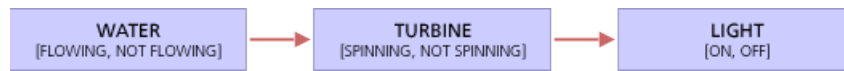
4.2.2 Module 4 | Causal Chains and Direct vs Indirect Causation

If one variable only influences another through some intermediate variable, then there is no arrow between the first feature and the third feature in the chain. For example, in a scenario involving a light being powered by hydro-electricity, the amount of water coming from the dam through the spout influences the speed of the turbine which influences whether electricity is generated to power the light bulb:

The water is still a cause of the light bulb, but only an indirect one. If the variables and their values are:

Variables For Turbine System	
Variable	Value
WATER	[Flowing, Not flowing]
TURBINE	[Spinning, Not spinning]
LIGHT	[On, Off]

then the causal graph is as follows:



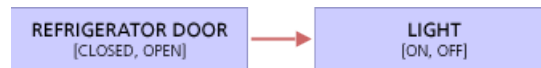
In the following two exercises, you are asked to construct a causal chain and a graph that has both an indirect and direct cause.

Causal chains highlight the fact that the idea of direct causation only makes sense relative to the set of variables under consideration. In the causal graph below, for example, the variable REFRIGERATOR DOOR is a direct cause of the variable LIGHT SWITCH, and the variable LIGHT SWITCH is a direct cause of the variable REFRIGERATOR LIGHT, but the state of the REFRIGERATOR DOOR is not a direct cause of the state of the REFRIGERATOR LIGHT relative to the system : {REFRIGERATOR DOOR, LIGHT SWITCH, REFRIGERATOR LIGHT}.



Why? Because if we fix the variable LIGHT SWITCH at either of its values, then bringing about a change in the state of the REFRIGERATOR DOOR will have no influence on the REFRIGERATOR LIGHT.

If we were only discussing the system: {REFRIGERATOR DOOR, REFRIGERATOR LIGHT}, then the door is a direct cause of the light:



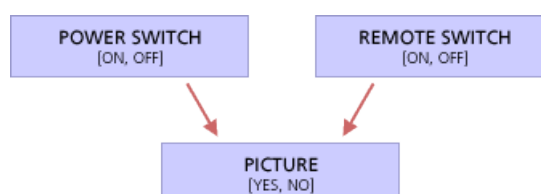
4.2.3 Module 4 | Common Effects

A variable E is a common effect of two or more variables X and Y when both X and Y are both causes of E, and at least one causal path from X to E does not involve Y, and at least one causal path from Y to E does not involve X.

For example, consider the following three variables applied to TVs that function normally.

Variables For A Tv System	
Variable	Value
POWER SWITCH (on the TV)	[On, Off]
REMOTE SWITCH	[On, Off]
PICTURE	[Yes, No]

The causal graph for these variables is:



Here PICTURE is a common effect of both REMOTE SWITCH and POWER SWITCH. This is a case in which each of the “cause” variables independently causes the common effect.

When we have interacting causes, the graph is the same in structure. So the structure of the causal graph is by no means completely informative about the nature of the causal relationship. Consider the following set of variables:

Variables For Agriculture System

Variable	Value
WATER (did the crops get rain)	[On, Off]
FERTILIZER (did the fertilizer get water)	[Yes, No]
GROWTH (did the plants grow well)	[Yes, No]

Because plants grow well only if they are *both* watered and have fertilizer, we say the causes interact to cause GROWTH. learn by doing

4.2.4 Module 4 | Cyclic Causal Graphs

Causation among variables is asymmetric. That is, if X is a cause of Y, then it doesn’t follow that Y is a cause of X. “Is a sibling of” is an example of a symmetric relationship. An example of an asymmetric relationship among people is “likes.” Causation among variables is asymmetric, but it isn’t antisymmetric. A relationship is antisymmetric if the fact that it holds one way precludes it holding the other. For example, the relationship “is a parent of” is antisymmetric. If person X is a parent of Y, then Y cannot be a parent of X.

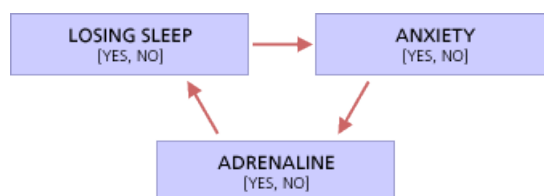
It is possible for one variable X to be a cause of Y and also for Y to be a cause of X. For example, losing sleep can cause anxiety, and anxiety can also cause a loss of sleep. Higher wages can cause inflation, and inflation can cause higher wages. Success causes confidence, and confidence causes success. In each of these cases, we say that there is a *direct cycle* in the causal graph.



What does it mean to say that there is a direct cycle between SUCCESS and CONFIDENCE? We can just apply the definition we have given for a “direct cause” twice:

- SUCCESS → CONFIDENCE: There are test pairs of causal assignments for SUCCESS that make a difference to the probability over CONFIDENCE.
- CONFIDENCE → SUCCESS: There are test pairs of causal assignments for CONFIDENCE that make a difference to the probability over SUCCESS.

So, a direct cycle is just when we have two variables, X and Y, and X is a direct cause of Y, and Y is also a direct cause of X. Cycles of causality need not be direct. For example, in the system including the variables LOSING SLEEP and ANXIETY, we might also include the variable: ADRENALINE. The system would now best be represented by the following causal graph, where the effect of ANXIETY on LOSING SLEEP is now indirect:



4.2.5 Module 4 | Case Studies

Causal graphs provide a qualitative but powerful representation of causal hypotheses. Hopefully, they will prove useful in understanding the causal and statistical reasoning discussed routinely in the newspaper. In analyzing these sorts of studies, you need to be able to at least:

1. identify the variables discussed in the study, and
2. draw the causal graph (or graphs) that represents the hypotheses under discussion in the study.

In this section we demonstrate how we would do it and then ask you to try. We start with a simple case, and then examine one that is slightly more complicated.

First, consider a case study on how using night lights might cause nearsightedness (myopia).

Case Study: New Study Vindicates Night Lights

Author: Eric Nagourney

Date: May 22, 2001

Source : <http://www.nytimes.com>

Copyright: 2001 New York Times Company

Concepts

- variables
- causal graphs
- confounders
- interventions
- experiments

Keywords

- night lights
- myopia

The night light has long been a source of comfort for the young, to say nothing of the guidance it provides those adults who would otherwise stumble through the dark. But some researchers had theorized that the lights might cause nearsightedness in infants.

Now, a new study reports that the concern appears unfounded. Writing in this month's *Investigative Ophthalmology and Visual Science*, scientists say they can find no link between myopia, the medical term for nearsightedness, and the lights.

The researchers, Dr. Dolores V. Bradley of Emory University and Dr. Earl L. Smith III of the University of Houston, suggested another possible explanation for the findings of a 1999 study reporting a higher incidence of nearsightedness among children who slept with night lights or overhead lights before age 2.

Myopic parents are more likely to have myopic children, the researchers said. And such parents may be more likely to use night lights, leading to a possible misinterpretation, the report said.

The more recent conclusions were based on a study of infant rhesus monkeys at the Yerkes Regional Primate Research Center at Emory. The researchers exposed nine monkeys to constant light for six months and then compared their eyesight with that of 23 monkeys raised in normal light conditions. Using mechanical means to assess sight, the researchers found no difference between the sets of monkeys.

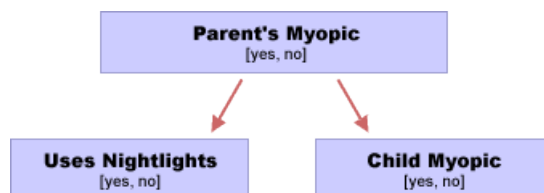
The 1999 study that found a link between myopia and night lights was based on the recollections of parents with nearsighted children.

There are three main variables discussed in the study:

- Nightlights before 2: [yes, no]

- Myopic: [yes, no]
- Myopic Parents: [yes, no]

Having identified the variables, what are the causal graph or graphs that the authors discuss connecting these variables? First, the scientists report their belief that nightlight use has no effect on myopia, as previously thought. They hypothesize that the association found in a previous study between nightlight use and myopia might be explained by a common cause: having myopic parents. It is certainly the case that the genes that make one's parents myopic get passed on to children – so a parent's myopia is a cause of a child's myopia. The authors speculate that parents who are myopic are more likely to use nightlights. Thus a plausible causal graph expressing the article's findings is the following:



Now read the next case study on Fetal Alcohol Syndrome and Mental Illness.

Case Study: Fetal Exposure to Alcohol Linked to Mental Illness

Date: Alcoholism & Drug Abuse Week; May 13, 1998

Source : <https://www.mylifepath.com/article/iac/45901>

Copyright: 1998 Manisses Communications Group

Concepts

- variables
- relative frequencies/associations
- causal graphs
- confounders

Keywords

- pregnancy
- alcohol
- mental illness

According to a study by researchers at the University of Washington School of Medicine, exposure to alcohol while in the womb increases the risk of developing a mental illness. Studying 25 adults who had been exposed to alcohol during their mothers' pregnancy, researchers found that 18 had received psychiatric treatment at some point in their lives. Fifteen of the subjects reported alcohol or drug dependence, 11 had experienced a major depressive episode and 10 reported a psychotic disorder.

Reports of anxiety disorders and eating disorders also were common in the group. "These results indicate that individuals with fetal alcohol syndrome or fetal alcohol effects manifested clinically significant mental illness as they matured," the authors wrote in the April issue of the *American Journal of Psychiatry*. Researchers added that the small sample size in this study necessitates further research into the link between fetal alcohol exposure and mental illness.

Consider a slightly more complicated case study on how Smoking causes the skin to age prematurely.

Case Study: Smoking linked to premature ageing

Date: April 12, 2000

Source : <http://news6.thdo.bbc.co.uk/low/english/health/default.stm>

Copyright: 2000 BBC

Concepts

- variables
- causal graphs
- interventions
- experiments

Keywords

- smoking
- skin
- ageing

The quick way to age Smoking destroys the ability of the skin to renew itself effectively, thus accelerating the ageing process, a study has indicated.

It has long been thought that smoking ages the skin prematurely and leads to the formation of more wrinkles. Now a team of researchers from Nagoya City University Medical School in Japan says it has discovered the reasons why smoke has such a damaging effect.

They found that cells exposed to smoke produced far more of the enzyme responsible for breaking down skin. Skin stays healthy and young-looking because it has the ability to constantly regenerate itself. This process depends on a subtle balance between the ability to break down old skin, and to create fresh replacement tissue. The body breaks down old skin with enzymes called matrix metalloproteinases (MMPs). These enzymes chop up the fibres that form collagen – the tissue that makes up about 80% of normal skin. New Scientist magazine reports that the researchers tested the impact of cigarette smoke by pumping it through a saline solution and adding the resulting mixture to collagen-producing skin cells. They found that cells exposed to smoke produced far more MMP than normal skin cells.

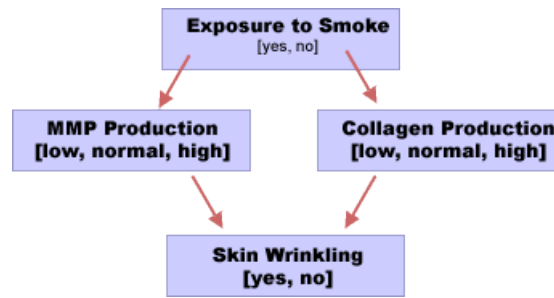
The researchers also found that smoke caused a drop in the production of fresh collagen by up to 40%. Lead researcher Dr Akimichi Morita said the combined effect is probably what causes premature skin ageing in smokers. His team found the effect was more pronounced when cells were exposed to more concentrated cigarette smoke. Dr Morita said: “This suggests the amount of collagen is important for skin ageing. It looks like less collagen means more wrinkle formation.” Dr Morita plans to confirm his findings by testing skin samples from smokers and non-smokers of various ages to see if the smoking has the same effect on collagen. “So far we have only done this in the lab. We don’t know exactly what happens in the body yet – that might take some time.”

Dr Lawrence Parish, director of the Centre for International Dermatology at Thomas Jefferson University Hospital in Philadelphia said the study confirmed what has long been expected. “Tobacco smoke is injurious to skin.” Amanda Sandford, of the anti-smoking charity Action on Smoking and Health (ASH), said the research might help to make young people think twice about taking up the habit. She said: “This will be useful in health education campaigns, explaining to people the effect smoking has on the skin. In particular, it might have more of an impact on young people who are often more concerned with their appearance than they are about their internal organs.”

Lots of concepts are discussed, but there are four main variables studied in the Nagoya City Lab:

- Exposure to Smoke: [yes, no]
- MMP Production: [low, normal, high]
- Collagen Production: [low, normal, high]
- Skin Wrinkling: [yes, no]

Having identified the variables, what is the causal graph that the authors believe holds over these variables? First, although the article takes it for granted that Exposure to Smoke is a cause of Skin Wrinkling, it seeks to elucidate the mechanisms by which this happens, and thus includes the variables MMP Production and Collagen Production. By adding these variables to the system, Exposure to Smoke is no longer a direct cause of Skin Wrinkling. The direct causes are rather MMP Production and Collagen Production. Do MMP Production and Collagen Production cause each other? The article doesn’t make it clear, but it seems to imply that the mechanisms by which Smoking increases MMP Production and decreases Collagen Production are independent. Thus, a plausible causal graph expressing the article’s findings is the following:



Now read the next case study on Exercise and Halting Mental Decline.

Case Study: Exercise ‘could halt mental decline’

Date: January 16, 2001

Source : unknown

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Concepts

- variables
- causal graphs
- interventions
- experiments

Keywords

- exercise
- mental decline

Exercise can improve mental abilities in the elderly, and go some way to preventing mental decline, say researchers. Scientists found mental abilities, or cognitive functioning, improved with exercise.

The reason for the link is unclear, but the researchers believe it could be linked to the fact exercise improves the body’s ability to pump blood and blood’s oxygen-carrying capacity.

The findings were a side benefit of research comparing the effects of exercise and drugs in fighting depression. Improvements were seen “above and beyond” what was expected after the depression had lifted, say researchers.

James Blumenthal, a psychologist at the Duke University Medical Center, in Durham, North Carolina, said one of the key findings was that exercise had beneficial effects on functions controlled by specific areas of the brain.

Memory, planning, organisation and juggling different tasks all improved under the study.

But attention, concentration and psychomotor skills, controlled by different parts of the brain, were not affected.

Exercise benefits

The research, named Smile (Standard Medical Intervention and Long Term Exercise) involved 156 patients aged between 50 and 77, who had a major depressive disorder.

They were randomly assigned to three groups. One did 30 minutes of aerobic exercise – using an exercise bike, walking or jogging – three times a week, the second group took an anti-depressant, and the third a combination of the two.

After 16 weeks, all three groups showed the same degree of improvement in standard measures of depression, leading the researchers to conclude exercise was as effective as medication.

In addition to testing levels of depression, the scientists also measured mental processes, comparing the 42 people in the exercise group with the 42 in the medication group.

They found exercise appeared to benefit some areas of cognitive function. Patients with milder depression showed the most improvement.

‘Offset decline’

Dr Blumenthal, who led the research, published in the Journal of Ageing and Physical Activity, said: “The implications are that exercise might be able to offset some of the mental declines that we often associate with the ageing process.”

“Further studies are warranted not only to clarify specific mental processes that are improved by exercise, but to better understand the underlying mechanisms of these improvements.”

He added: “It is thought that one of the reasons why the elderly – especially those with coronary artery disease or hypertension – tend to suffer some degree of cognitive decline is in part due to a reduction in blood flow to the brain.”

“So it may be that just as exercise improves muscle tone and function, it may have similar effects on the brain.”

A spokeswoman for Age Concern said “Exercise is extremely beneficial to older people, both in terms of getting out and about and meeting new people, and obviously the health related benefits of keeping mobile and better fitness in general.”

4.3 Summary

This module stressed four basic skills and/or concepts:

1. Elements of Causal Graphs
2. Causal Graphs and Response Structures
3. Representing Different Varieties of Causation as Graphs
4. Translating Text into Causal Graphs

Causal graphs provide a powerful qualitative representation of causal relations among variables. A causal graph includes a set of variables, and a set of directed edges that connects pairs of these variables. The edges are “directed” because causation is asymmetric and has a direction. If one variable X is a direct cause of another variable Y in some causal system S , then we include an arrow, or directed edge, from X to Y in the causal graph that represents S .

What does it mean for one variable to be a direct cause of another relative to a system of variables S , that is, for there to be an arrow from X to Y in the causal graph over S ? The existence of an arrow corresponds to whether or not the relationship between X and Y satisfies the definition we gave using response structures and test pairs in modules 1.2 and 1.3. X is a direct cause of Y relative to a set of variables S just in case there are test pairs of causal assignments for X across which there is a difference in Y . There is an arrow from X to Y in the causal graph for S if and only if X is a direct cause of Y according to this definition.

If there is chain of arrows from A to D , i.e., $A \rightarrow B \rightarrow C \rightarrow D$, etc., then we say there is a *causal chain* from A to D , and A is an indirect cause of D . We also say A is a cause of D . If A is a cause of B , and A is a cause of C , then we say A is a common cause of B and C . Although causation among variables is asymmetric, it is not anti-symmetric, so it is possible for one variable A to be a cause of B and also for B to be a cause of A . In such a case, we say the causal graph has a *direct cycle*. If there is a chain of edges leading from any variable back to itself, then we say the graph has a *cycle*. If a graph has a cycle, we say it is a *cyclic graph*. If the graph has no cycle, then we say it is *acyclic*.

Whether or not there is an edge from one variable to another depends on what other variables we have included in the system. If, for example, there is a causal chain $A \rightarrow B \rightarrow C$, but A has no direct influence on C that doesn’t go through B , then we don’t include an edge from A to C . If we consider a system that doesn’t include B , however, then relative to that system, we need to include an $A \rightarrow C$ edge.